TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	FuSaGuard: ML-Enabled Mission-Mode Functional Safety for ADAS Compute SoCs			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical and Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Service and application security	Home Appliances and Electronic Application Devices	Other Home Appliances and Electronic Application Devices
Concept of Proposed Technology	**Problem Statement and Market Need** ADAS and autonomous driving platforms increasingly rely on complex, high-performance compute SoCs that must operate safely over long lifetimes under automotive stress conditions. As design complexity and integration density increase, safety-critical failures increasingly arise not from design-time bugs alone, but from **latent defects, transient faults, and aging-related degradation that manifest during mission-mode operation**. As per ISO 26262, Functional Safety is an approach to ensure automotive electronics fail safely, without causing any safety violations to humans as well as equipment.[1, 2] Current functional safety (FuSa) validation approaches face three major limitations: 1. **High cost and poor scalability** of exhaustive fault injection and simulation campaigns. 2. **Dependence on intrusive test modes** that are unavailable or undesirable in deployed systems. 3. **Limited observability and diagnosability**, particularly in AMS blocks such as clocking, power management, and data conversion circuits. These limitations directly impact Korean SMEs developing ADAS SoCs, IP blocks, validation tools,			

	or safety solutions, increasing development cost and delaying commercialization.
	Core Technology Description
	FuSaGuard is a unified, ML-enabled framework for **mission-mode functional safety monitoring** of ADAS compute SoCs, spanning both digital logic and analog/mixed-signal (AMS) subsystems. The framework integrates four tightly coupled capabilities:
	1. **Digital Fault Criticality Analysis**
	Learning-based models analyze SoC structure, workloads, and sampled fault behavior to rank components by their contribution to system-level safety risk. This enables **targeted validation and monitoring**, replacing costly exhaustive fault injection with guided, risk-aware analysis.
	2. **Non-Intrusive Functional Test Generation**
	FuSaGuard generates design-aware functional tests that operate exclusively through existing interfaces and realistic workloads. This enables safety monitoring **without dedicated test modes**, preserving availability and enabling deployment in field-like conditions.
	3. **Runtime Detection and Diagnosis**
	Lightweight runtime monitors analyze execution traces using contrastive and representation-learning techniques to rapidly distinguish fault-free and faulty behavior, enabling **early detection and localization** of safety-critical deviations during operation.
	4. **AMS Functional Safety with Explainability**
	Unsupervised anomaly detection methods monitor AMS components under realistic process, voltage, temperature (PVT), and environmental variation. Explainable outputs provide **ranked root-cause signals**, allowing engineers to differentiate true safety violations from benign drift.
	Together, these capabilities enable continuous, explainable, and cost-effective safety assurance during normal operation—an increasingly important requirement for next-generation ADAS platforms.
	Deployable Outcomes for Korean SME Partners
	Through the joint R&D project, the Korean SME partner will obtain:
	• A **prototype FuSaGuard software toolkit** that outputs:
	  ○ Ranked safety-critical SoC components
	  ○ Targeted functional tests
	  ○ Lightweight runtime safety monitors
	• **Benchmark workloads, fault models, and evaluation scripts** tailored to ADAS SoCs, usable for:
	  ○ Internal safety qualification
	  ○ Customer demonstrations
	  ○ Certification preparation
	• **ISO 26262-aligned integration and reporting guidance**, enabling rapid incorporation into existing verification, validation, and post-silicon workflows.
	These outcomes are designed for **direct adoption or commercialization** by Korean SMEs as internal tools, licensable modules, or integrated offerings.

Rationale for Joint R&D Project	**Why a Joint R&D Project Is Necessary** While the core concepts of **FuSaGuard** have been validated at an early research level, transitioning to an industry-ready solution requires: • Access to **realistic ADAS SoC designs and IP blocks** • **Automotive-grade operating constraints and workloads** • Integration with **industrial verification and post-silicon validation infrastructure** These elements are best provided by an industry partner. A joint R&D project enables rapid co-development, realistic evaluation, and early alignment with commercial requirements, ensuring the technology is practical and deployable. **Why Collaboration Is Essential for Commercialization** The target outcome of this project is a **partner-ready safety monitoring module** suitable for commercialization. Korean SME participation is essential to: • Ensure compatibility with existing toolchains and workflows • Validate performance on representative IP (CPU clusters, accelerators, interconnects, memory, AMS blocks) • Align outputs with customer and certification expectations • Define licensing, packaging, and go-to-market strategies This collaboration directly supports TIPA's goal of strengthening the **global competitiveness of Korean SMEs** in advanced mobility and cybersecurity markets.

Key Words	1. Functional Safety (ISO 26262) 2. ADAS compute SoC safety monitoring 3. Mission-mode testing and runtime assurance 4. ML-based fault detection and diagnosis 5. AMS anomaly detection and explainability

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (4)

R&D Objectives	**Project objective** Develop and validate, at **TRL 4**, a mission-mode functional safety monitoring and evaluation toolkit for ADAS compute SoCs that: • Reduces safety validation cost and time • Improves early detection and diagnosis of safety-critical faults • Produces outputs directly usable as ISO 26262 safety evidence **Specific R&D objectives** • Rank SoC components by safety relevance using ML-based fault criticality modeling[4] • Generate non-intrusive, design-aware functional tests through existing interfaces[3] • Develop lightweight runtime monitors with minimal performance and memory overhead[5] • Detect and explain AMS safety violations under realistic PVT variation[6] • Establish repeatable benchmarks and metrics suitable for partner adoption and certification support
R&D Target Specifications	**Target specifications for TRL 2 → TRL 4 prototype validation** • Digital criticality accuracy: identify the top safety-critical 10% of components with ≥90% agreement against fault-injection-based ranking on representative SoC modules • Acceleration: reduce required fault-injection simulations by 5×–10× via guided sampling while maintaining comparable safety-risk ranking quality • Mission-mode test efficiency: achieve ≥3× higher detection rate per test vector than unguided functional testing, with no dedicated scan mode requirement • Runtime overhead: demonstrate monitor overhead under 5% compute and under 2% memory for representative workloads on a SoC-level simulation or emulation environment. • Detection responsiveness: flag safety-critical deviations within 10–100 ms after manifestation for representative ADAS workloads, adjustable to partner constraints • AMS anomaly detection: ≥90% recall on seeded anomaly classes under realistic PVT variation, with ≤5% false alarm rate after calibration • Explainability: for each flagged event, output a ranked list of likely root-cause signals or blocks (top-5) that engineers can act on
R&D Plan	**Overall approach** The joint R&D is structured into **three technical work packages** and **one transfer/commercialization work package**, aligned with TIPA's two-phase program structure: • **Phase 1 (6-month Pre-Study):** Feasibility demonstration, planning, early prototypes, and commercialization alignment. • **Phase 2 (2.5-year Main Research):** TRL-4 prototype development, SoC-level validation, benchmarking, and transfer-ready packaging. **<u>Work Package 1 (WP1): Digital Functional Safety for ADAS Compute SoCs</u>** WP1 Objective (Evaluator-Facing) Develop deployable, ML-based digital functional safety techniques that enable Korean SMEs to **prioritize safety effort, reduce fault-injection cost, and improve diagnosability** of ADAS compute SoCs during mission-mode operation. *<u>WP1.1 – Data and Fault Model Definition</u>* Define representative fault models, safety metrics, and ADAS workloads in close collaboration with the SME partner, using real design artifacts and validation constraints. **SME Value:** Ensures that all subsequent methods are aligned with **partner-relevant safety risks**, reducing the gap between research results and industrial adoption. *<u>WP1.2 – ML-Based Digital Fault Criticality Analysis</u>* Develop machine-learning models that learn component-level safety criticality using design structure, workload behavior, and sampled fault-injection data.

SME Value:
Enables SMEs to **replace exhaustive fault-injection campaigns** with risk-aware prioritization, significantly reducing simulation cost, time-to-safety-evidence, and compute resource usage.
WP1.3 – Non-Intrusive Functional Test Generation
Generate targeted functional tests using reinforcement learning and design-aware constraints that operate exclusively through existing functional interfaces.
SME Value:
Allows safety validation **without scan or special test modes**, making the approach suitable for post-silicon validation, field testing, and customer demonstrations.
WP1.4 – Runtime Detection and Diagnosis
Develop lightweight runtime monitors based on contrastive learning over execution traces to rapidly detect and localize safety-critical deviations.
SME Value:
Provides **early warning and fault localization during real operation**, improving field reliability and reducing recall exposure for ADAS products.
Work Package 2 (WP2): AMS Functional Safety for ADAS Platforms
WP2 Objective (Evaluator-Facing)
Enable practical functional safety monitoring of safety-critical AMS blocks, which are currently under-served by digital-only safety methodologies.
WP2.1 – Definition of AMS Safety Violation Signatures
Identify safety-relevant signals, violation signatures, and metrics for AMS components such as PLLs, clocking networks, ADCs, and power management interfaces.
SME Value:
Directly addresses **one of the weakest points in current ADAS safety flows**, improving coverage where failures are costly and difficult to diagnose.
WP2.2 – Unsupervised AMS Anomaly Detection
Develop unsupervised anomaly detection techniques robust to noise, process variation, temperature, and environmental uncertainty.
SME Value:
Eliminates dependence on large labeled fault datasets, making the solution **practical for SMEs** with limited historical failure data.
WP2.3 – Explainable Root-Cause Analysis for AMS Faults
Develop explainability and causal inference methods that output ranked root-cause signals and blocks for each detected anomaly.
SME Value:
Transforms raw anomaly alerts into **actionable engineering insight**, reducing debug time and increasing trust in ML-based safety tools.
Work Package 3 (WP3): SoC-Level Integration and Benchmarking
WP3 Objective (Evaluator-Facing)
Demonstrate that FuSaGuard operates as a **unified, scalable safety solution** at the SoC level and produces outputs usable as safety evidence.
WP3.1 – Unified Digital + AMS Integration
Integrate digital and AMS safety monitors into a single end-to-end safety monitoring pipeline for a representative ADAS compute SoC use case.
SME Value:
Reduces integration burden and enables **system-level safety visibility**, aligning with how ADAS products are evaluated by customers and certifiers.
WP3.2 – Benchmark Suite and Evaluation Framework
Develop reusable datasets, fault scenarios, workloads, and automated evaluation scripts with metrics for coverage, latency, overhead, and robustness.
SME Value:
Provides SMEs with a **repeatable evaluation framework** that can be reused across projects and customers, improving consistency and credibility.
WP3.3 – ISO 26262 Safety Evidence Mapping
Translate monitoring outputs into ISO 26262-aligned artifacts such as diagnostic coverage estimates, monitor effectiveness metrics, and traceability reports.
SME Value:
Directly supports **certification and customer audits**, reducing the cost and uncertainty of preparing safety cases.
Work Package 4 (WP4): Technology Transfer and Commercialization
WP4 Objective (Evaluator-Facing)
Ensure that FuSaGuard transitions from a research prototype into a **commercially viable, partner-ready product or module**.

WP4.1 – Integration into Partner Workflow
Align FuSaGuard APIs, data formats, and reports with the partner's verification, validation, and post-silicon workflows.
SME Value:
Minimizes adoption friction and enables **rapid internal deployment** within existing toolchains.
WP4.2 – Market Analysis and Pilot Planning
Define target customer segments, competitive positioning, pilot success metrics, and entry strategies across ADAS SoC vendors, Tier-1 suppliers, EDA vendors, and mixed-signal IP providers.
SME Value:
De-risks commercialization by aligning technical development with **clear market demand and revenue pathways**.
WP4.3 – IP Strategy and Phase-2 Commercial Roadmap
Define background and foreground IP, licensing intent, disclosure boundaries, and partner-led go-to-market strategy.
SME Value:
Provides clarity on **IP ownership and monetization**, enabling confident investment in productization and sales.
Timeline and milestones aligned to TIPA phases
Phase 1, Pre-study and evaluation, July 1 to December 31, 2026
Phase 1 Objective:
Establish technical feasibility, confirm partner alignment, and define a scalable R&D and commercialization plan based on representative ADAS SoC use cases.
Key Activities:

- Finalize the detailed Phase-2 R&D plan, including task breakdowns, datasets to be created, evaluation metrics, and integration interfaces.
- Define partner-relevant safety metrics, fault models, and representative ADAS workloads.
- Demonstrate technical feasibility on representative blocks:
  - Digital fault criticality prototype
  - Initial AMS anomaly detection module
  - Minimal end-to-end safety monitoring pipeline on a representative ADAS SoC slice
- Conduct market analysis and IP discussions to identify:
  - Commercialization routes
  - Tool integration points
  - Partner-specific adoption and pilot plans

Phase 1 Milestones and Deliverables (by December 31, 2026):

- Approved **Phase-2 detailed R&D execution plan**
- **Pre-study evaluation report** documenting feasibility results
- **Prototype v0 demonstrations** for digital and AMS safety monitoring
- **Market analysis and IP strategy report**, including preliminary commercialization pathway

TIPA Evaluation Relevance:
Phase 1 de-risks the project by validating feasibility early and ensuring that Phase-2 research is tightly aligned with SME needs and commercialization goals.
Phase 2, Main research and project management, January 1, 2027, to June 30, 2029
Phase 2 Objective:
Develop, integrate, and validate a **TRL-4 prototype** of FuSaGuard, and deliver a **transfer-ready safety monitoring toolkit** suitable for commercialization or internal deployment by the Korean SME partner.
2027 – Core Technology Development
2027 H1

- Digital fault criticality model v1
- Sampled fault-injection and guided validation workflow
- AMS anomaly detection v1 with robustness checks under PVT variation

2027 H2

- Non-intrusive functional test generation v1
- Lightweight runtime detection and diagnosis monitors
- Initial explainability and root-cause ranking outputs

Milestone:
Validated subsystem-level safety monitoring for both digital and AMS components.
2028 – SoC-Level Integration and Safety Evidence
2028 H1

- Full SoC-level integration of digital and AMS monitoring

	• Benchmark suite v1 including datasets, workloads, and fault scenarios
	• Automated evaluation scripts for coverage, latency, overhead, and robustness
	2028 H2
	• ISO 26262-aligned safety evidence templates
	• TRL-4 laboratory-scale validation on partner-agreed workloads
	• Validation against partner-defined performance and overhead budgets
	Milestone:
	Demonstrated system-level functional safety monitoring with outputs directly usable as safety evidence.
	2029 – Technology Transfer and Commercialization Readiness
	2029 H1
	• Transfer-ready FuSaGuard toolkit
	• Stable APIs, documentation, and reporting formats
	• Final benchmark suite and evaluation framework
	• Partner pilot evaluation and feedback
	• Final commercialization package and licensing plan
	Final Deliverables:
	• TRL-4 validated FuSaGuard prototype
	• Integration-ready software toolkit
	• Benchmark and evaluation assets
	• Commercialization and IP roadmap for partner-led deployment
	Project Management and Review Structure
	• Quarterly **joint steering committee meetings** to review milestones, risks, and alignment with commercialization objectives.
	• Monthly technical synchronization meetings and shared issue tracking.
	• Structured risk management focused on:
	  ○ Data availability
	  ○ Runtime overhead constraints
	  ○ False alarm and robustness control
	• Coordinated IP management documenting background and foreground IP while meeting TIPA disclosure requirements.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will lead the **core technical innovation and prototype development** of FuSaGuard, ensuring scientific rigor, methodological soundness, and TRL progression. RPI's responsibilities include: • **Core algorithm and model development** Design and implement ML-based methods for digital fault criticality analysis, non-intrusive functional test generation, and runtime detection and diagnosis suitable for mission-mode operation. • **AMS functional safety innovation** Develop unsupervised anomaly detection techniques for AMS blocks, including signal selection, feature extraction, robustness to PVT variation, and explainable root-cause analysis outputs usable by practicing engineers. • **Integrated prototype toolkit development** Build, maintain, and document the FuSaGuard prototype, including modular APIs, configuration interfaces, and reporting formats aligned with industrial workflows. • **Benchmarking and evaluation methodology** Design benchmark datasets, fault scenarios, workloads, and evaluation protocols, and generate quantitative evidence for detection accuracy, latency, overhead, robustness, and scalability. • **Technology transfer support** Support partner adoption through documentation, reference implementations, and technical training, and co-develop Phase-2 requirements for productization and commercialization. **TIPA Alignment:** RPI's role ensures that the project delivers **technically differentiated, globally competitive technology**, while systematically transferring know-how to the Korean SME partner.

Expected R&R of Korean SME Partner	The Korean SME partner will lead **industrial integration, validation realism, and commercialization**, ensuring that FuSaGuard evolves into a deployable and market-relevant solution. The SME partner's responsibilities include: • **Provision of representative industrial artifacts** Provide access to representative ADAS SoC design artifacts, IP models, or abstractions, and define mission-mode safety requirements, constraints, and performance budgets. • **Industrial validation infrastructure and workflows** Enable evaluation within partner validation environments, including simulation regressions, emulation, post-silicon debug flows, and safety documentation practices. • **Integration and usability evaluation** Lead integration of FuSaGuard into partner workflows, providing feedback on usability, scalability, performance, and alignment with real engineering decision-making. • **Benchmark and metric validation** Contribute realistic workloads and fault scenarios and validate that FuSaGuard outputs translate into actionable safety insights and certification-relevant metrics. • **Commercialization leadership** Lead market exploration, customer discovery, pilot engagement, and go-to-market planning across target segments, including ADAS SoC vendors, Tier-1 suppliers, EDA vendors, and mixed-signal IP providers. **TIPA Alignment:** The SME partner's leadership ensures **commercial pull, rapid adoption, and sustainable economic impact**, directly supporting TIPA's mission to strengthen Korean SME competitiveness.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected technical outcomes** By the end of the project, the joint R&D effort will deliver: • A **TRL-4 validated FuSaGuard prototype** enabling mission-mode functional safety monitoring across both digital and AMS components of ADAS compute SoCs. • Quantified **acceleration of safety validation**, achieved through criticality-guided fault injection and targeted functional testing, with measured reductions in validation cost and time. • A robust **AMS anomaly detection and explainability module** capable of early detection of safety-critical deviations under realistic operating variation. • A reusable **benchmark suite, datasets, and evaluation scripts** that enable repeatable assessment and partner adoption. • **ISO 26262-aligned reporting templates and safety evidence artifacts**, supporting certification preparation and customer audits. These outcomes collectively demonstrate a practical, system-level approach to functional safety assurance beyond traditional intrusive testing methods. **Commercialization potential and industry impact** FuSaGuard addresses a well-defined and growing industry need: generating credible functional safety evidence for increasingly complex ADAS compute SoCs without relying solely on costly exhaustive testing or intrusive test modes. **Commercialization pathways for the Korean SME partner include:** • Deployment as an **internal productivity tool** to reduce validation cost and time-to-certification. • Packaging as a **licensable software module** for ADAS SoC vendors or Tier-1 suppliers. • Integration as a **value-added feature** within EDA, validation, or post-silicon debug platforms. • Bundling with **mixed-signal IP offerings** to provide differentiated safety analytics. **Value to Industry Stakeholders:** • **SoC design houses:** earlier identification of high-risk design regions and reduced safety-case preparation time. • **Tier-1 ADAS and ECU suppliers:** improved field reliability and reduced recall exposure. • **EDA and test-tool vendors:** differentiated functional safety analytics capabilities. • **Mixed-signal IP vendors:** enhanced diagnosability and certification readiness.

6. References

[1] ISO 26262, Road vehicles - Functional safety.

[2] IEC 61508, Functional safety of electrical/electronic/programmable electronic safety-related systems.

[3] Sanjay Das, Swastik Bhattacharya, Anand Menon, Shamik Kundu, Pooja Madhusoodhanan, Prasanth Viswanathan Pillai, Rubin Parekhji, Arnab Raha, Suvadeep Banerjee, Suriya Natarajan, Kanad Basu. Machine Learning-Driven STL Generation for Enhancing Functional Safety of E/E Systems, At ACM/IEEE Design Automation Conference (DAC), 2025.

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[5] Ayush Arunachalam, Sanjay Das, Monikka Rajan, Fei Su, Xiankun Jin, Suvadeep Banerjee, Arnab Raha, Suriyaprakash Natarajan, Kanad Basu. Enhanced ML-based Approach for Functional Safety Improvement in Automotive AMS Circuits. At International Test Conference (ITC), 2023.

[6] Ayush Arunachalam, Shamik Kundu, Fei Su, Suvadeep Banerjee, Robert Jin, Arnab Raha, Suriyaprakash Natarajan, Kanad Basu. Unsupervised Learning-based Early Anomaly Detection in AMS Circuits of Automotive SoCs. At International Test Conference (ITC), 2022.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization